

Subject Description Form

Subject Code	COMP5241
Subject Title	Software Engineering and Development
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: 1. provide sufficient insight into the software development environment; 2. provide detailed knowledge of the application of typical software engineering techniques; 3. provide appreciation of CASE tools.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) understand and apply fundamental as well as advanced and specialized software engineering knowledge for formulating models and solutions, addressing complex interdisciplinary issues with professionalism; b) analyze and solve software engineering problems through critical thinking, analytical thinking and creative thinking; c) design and evaluate systems/applications to satisfy various stakeholder's needs, and incorporate start-of-the-art academic advancements in the related design and/or evaluation, addressing complex interdisciplinary issues with professionalism; d) work and collaborate effectively in teams possibly with multi-disciplinary backgrounds as a leader or a member, fostering intercultural and interdisciplinary interactions; e) communicate demands, solutions, and state-of-the-art academic advancements effectively through effective written reports/documents, effective presentations and other means of verbal/written communications; f) uphold professional ethics, responsibilities, demonstrating critical awareness and integrity in making responsible decisions; and g) engage in life-long independent learning for professional development, continuously adapting and reflecting on personal growth.
Subject Synopsis/ Indicative Syllabus	• Overview of software engineering and development: Definition of software engineering, Software development lifecycles (e.g. Waterfall vs. Agile), CASE (Computer-Aided Software Engineering) tools. • Requirement Analysis and Software Design: Requirement elicitation techniques, Software requirements specification (SRS), User Stories, Object-oriented analysis and design, Software design principles and patterns • Software Implementation: Overview/review of typical object-oriented/structured programming language. Version control and collaboration tools, Unit Testing, Continuous

	Integration (CI) and Continuous Delivery, Containerized software development, Cloud-based software development, Microservices architecture • Software Testing and Deployment: Software testing techniques, Functional and nonfunctional testing, Deployment Strategies and Models.								
Teaching/Learning Methodology	Class activities including – lecture, assignment, quiz, group project, and tutorial/seminar (if applicable).								
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						
			a	b	c	d	e	f	g
	1. Exercises, Assignments, quizzes & projects	30	✓	✓	✓	✓	✓	✓	✓
	2. Final examination	70	✓	✓	✓		✓	✓	
	Total	100							
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: a) Assignments, quizzes, projects, and/or final exam questions will involve fundamental as well as advanced and specialized software engineering knowledge for formulating models and solutions, hence will fulfill the need to assess intended learning outcome a); b) Assignments, quizzes, projects, and/or final exam questions will involve analyzing and solving software engineering problems creatively, hence will fulfill the need to assess intended learning outcome b); c) Software requirement engineering related assignments, quizzes, projects, and/or final exam questions will involve designing and evaluating systems/applications to satisfy user needs and various requirements, hence will fulfill the need to assess intended learning outcome c); d) Group project will fulfill the need to assess intended learning outcome d); e) In-class presentation/quiz-answering, assignment report writing, and final exam writing will fulfill the need to assess intended learning outcome e). f) Quizzes and/or final exams will cover ethics issues, hence will fulfill the need to assess intended learning outcome f); g) Assignments and projects will teach the students to learn by themselves and by teamwork, so as to fulfill the need to assess intended learning outcome g).								
Student Study Effort Expected	Class contact:								
	Class activities (lecture, quizzes, project presentation, tutorial/seminar (if applicable))		39Hrs.						
	Other student study effort:								

	Exercises, Assignment, group project, exams.	66Hrs.
	Total student study effort:	105Hrs.
Reading List and References	(1) Sommerville, I., 2015, Software Engineering, 10th ed., Pearson. (2) Pressman, R., Maxim, B., 2023, Software Engineering, A Practitioner's Approach, 9th ed., McGraw Hill. (3) Bennett, S., Ray, F., 2010, Object-Oriented Systems Analysis and Design Using UML, 4th Ed., McGraw-Hill.	